

# Solution to Question 5: Algorithm for Allocating $n \cdot k$ Items

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## Proposed Algorithm:

### Assumptions and Notation

- We are given  $n$  players and  $m = n \cdot k$  items.
- We assume the players have additive valuation functions.

### Algorithm Description

1. Fix an arbitrary ordering of the players:  $P_1, P_2, \dots, P_n$ .
2. Perform  $k$  rounds of selection.
3. In each round, every player (from  $P_1$  to  $P_n$ ) chooses their most preferred item from the set of remaining items.

## Proof of Correctness

### 1. Cardinality Constraint ( $k$ items per player)

In every round, each of the  $n$  players selects exactly one item. Since the algorithm runs for exactly  $k$  rounds, each player selects exactly  $k$  items. In total,  $n \cdot k$  items are distributed, as required.

### 2. Proof of EF1 (Envy-Free up to one item)

We need to prove that for any pair of players  $A$  and  $B$ , player  $A$  does not envy player  $B$  after removing one item from  $B$ 's bundle.

Let  $a_1, a_2, \dots, a_k$  denote the sequence of items chosen by player  $A$  (where  $a_i$  is chosen in round  $i$ ). Let  $b_1, b_2, \dots, b_k$  denote the sequence of items chosen by player  $B$ .

We distinguish between two cases based on the relative order of the players in the selection sequence:

**Case 1: Player  $A$  chooses before Player  $B$**  In every round  $i$ , when it is  $A$ 's turn to choose, item  $b_i$  (which  $B$  will choose later in the same round) is still available. Since  $A$  chooses the best item available to them at that moment, it holds that:

$$v_A(a_i) \geq v_A(b_i) \quad \forall i \in \{1, \dots, k\}$$

Summing over all items:

$$v_A(A) = \sum_{i=1}^k v_A(a_i) \geq \sum_{i=1}^k v_A(b_i) = v_A(B)$$

In this case,  $A$  does not envy  $B$  at all (EF), which implies EF1.

**Case 2: Player  $A$  chooses after Player  $B$**  In this case, it is possible that  $B$  selected an item that  $A$  preferred. However, observe the timing across rounds: When  $A$  chooses item  $a_i$  in round  $i$ , item  $b_{i+1}$  (which  $B$  will take only in the *next* round,  $i + 1$ ) is still available in the pool. Therefore,  $A$  could have chosen  $b_{i+1}$  but preferred  $a_i$ . This implies:

$$v_A(a_i) \geq v_A(b_{i+1}) \quad \forall i \in \{1, \dots, k-1\}$$

Now, let us verify the EF1 condition. We remove  $B$ 's first item,  $b_1$ , from their bundle. The value of  $B$ 's bundle without  $b_1$  according to  $A$  is:

$$v_A(B \setminus \{b_1\}) = \sum_{j=2}^k v_A(b_j) = \sum_{i=1}^{k-1} v_A(b_{i+1})$$

Comparing this to the value of  $A$ 's bundle:

$$v_A(A) = \sum_{i=1}^k v_A(a_i) \geq \sum_{i=1}^{k-1} v_A(a_i)$$

(The last inequality holds because valuations are non-negative). Using the inequality  $v_A(a_i) \geq v_A(b_{i+1})$  established earlier:

$$\sum_{i=1}^{k-1} v_A(a_i) \geq \sum_{i=1}^{k-1} v_A(b_{i+1}) = v_A(B \setminus \{b_1\})$$

**Conclusion:**

$$v_A(A) \geq v_A(B \setminus \{b_1\})$$

Thus, if we remove the first item chosen by  $B$ ,  $A$ 's envy is eliminated.

**Summary:** The Round Robin algorithm allocates exactly  $k$  items to each player and satisfies the EF1 property for every pair of players.

## Time Complexity

The time complexity of the allocation phase is  $O(n \cdot k)$ .

- We assume the input includes the sorted preference lists for each player.
- The algorithm proceeds in  $k$  rounds, with each of the  $n$  players performing one selection per round. Consequently, the total running time is  $O(n \cdot k)$ .